

## Experiment No: 1

### Aim: Gravimetric Estimation of Nickel by Dimethyl glyoxime

#### Theory:

Gravimetric analysis is one of the most accurate analytical methods available. It is concerned with the determination of a substance by the process of weighing. The element or radical to be determined is converted into a stable compound of definite composition and the mass of the compound is determined accurately. From this, the mass of element or radical is calculated.

The gravimetric analysis involves a) precipitation b) filtration c) washing of the precipitate d) drying e) weighing of the precipitate.

There are four types of gravimetric analysis:

1. Physical gravimetry
2. Thermogravimetry
3. Precipitative gravimetric analysis
4. Electrodeposition.

**Physical gravimetry:** Physical separation and classification of matter in environmental samples based on volatility and particle size (e.g., total suspended solids).

**Thermogravimetry:** Volatile solid samples are analyzed by this method. In this method the samples are heated and the changes in sample mass are recorded.

**Precipitative gravimetry:** The chemical precipitation of an analyte and weighing of the precipitate is done in the precipitative gravimetry.

**Electrodeposition:** It involves the electrochemical reduction of metal ions at a cathode and simultaneous deposition of the ions on the cathode.

#### Precipitative Gravimetric Analysis:

Precipitative gravimetric analysis requires that the substance to be weighed be readily removed by filtration. In order for a non-filterable precipitate to form, it must be supersaturated with respect to its solubility product constant. However, if it is too far above the saturation limit, crystal nucleation may occur at a rate faster than crystal growth (the addition of molecules to a crystal nucleus, eventually forming a non-filterable crystal). When this occurs, numerous tiny micro-crystals are formed rather than a few large ones. In the extreme case, micro-crystals may behave as colloids and pass through a fibrous filter. To avoid this, precipitating solutions may be heated. Because the solubility of most salts increases with increasing temperature, this treatment will lower the relative degree of super saturation and slow the rate of nucleation. Also, one might add the precipitant slowly with rapid mixing to avoid the occurrence of locally high concentrations.

### The Gravimetric Estimation of Nickel:

The nickel is precipitated as nickel dimethyl glyoxime by adding alcoholic solution of dimethyl glyoxime  $\text{C}_4\text{H}_6(\text{NOH})_2$  and then adding a slight excess of aqueous ammonia solution.

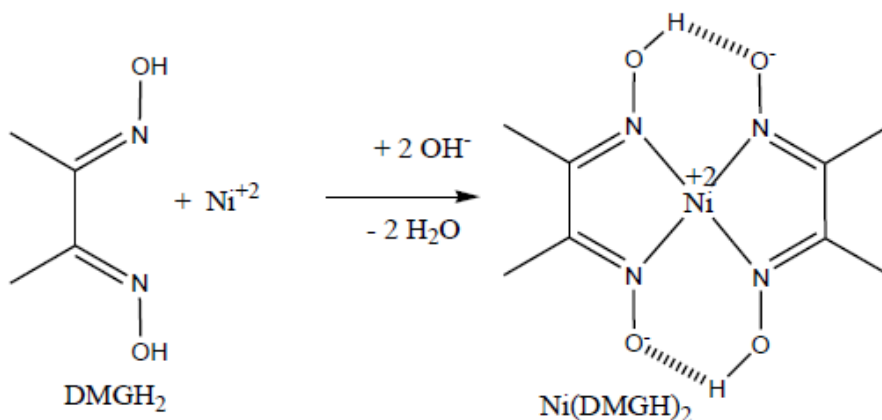


In the pH range 5 to 9, the formation of the red chelate complex occurs quantitatively in a solution. The chelation reaction occurs due to donation of the electron pairs on the four nitrogen atoms, not by electrons on the oxygen atoms. The reaction is performed in a solution buffered by either an ammonia or citrate buffer to prevent the pH of the solution from falling below 5. At very low pH reverse reaction takes place causing the hydrolysis of  $\text{Ni}(\text{DMGH})_2$  that leads to the formation of  $\text{Ni}^{+2}$  in the mother liquor.

A slight excess of the reagent has no action on the precipitate, but a large excess should be avoided because of the possible precipitation of the reagent itself. It is therefore crucial to avoid the addition of too large and excess of the reagent because it may crystallize out with the chelate. The nickel dimethylglyoximate is a very bulky precipitate. Therefore, the sample weight used in the analysis must be carefully controlled to allow more convenient handling of the precipitate during the transfer to the filtering crucible. The compactness of the precipitate is improved by adjusting the pH to 3 or 4, followed by the addition of ammonia solution.

A slow increase in the concentration of ammonia in the solution causes a slight increase in the pH gradually and results in the precipitation of the complex. The result is the formation of a denser precipitate. Once the filtrate has been collected and dried, the nickel content of the solution is calculated stoichiometrically from the weight of the precipitate.

*The structure of DMG & the complex with nickel ions is given below;*



**Procedure:**

- ☐ The given nickel solution is made up to 100mL in a standard measuring flask.
- ☐ 20 mL of solution is pipetted into a 250 mL beaker.
- ☐ About 5 mL 1:1 HCl is added and diluted to 150 mL.
- ☐ The solution is heated to 70-80°C. 25 mL of 1% dimethyl glyoxime in alcohol is added, immediately followed by dilute ammonia solution drop wise until it strongly smells of ammonia.
- ☐ The solution containing the precipitate is heated in a water bath for 30 minutes.
- ☐ The precipitate is allowed to stand for an hour.
- ☐ Filter the solution through a previously weighed sintered glass crucible.
- ☐ The precipitate is washed with cold water to free chloride.
- ☐ The crucible is placed in a dry 100 mL beaker and heated in the air oven at 110-120°C for 1 hour.
- ☐ It is cooled in a dessicator and weighed. Repeat drying until constant weight is obtained.

**Calculation:**

Mass of sintered glass crucible =  $x$  g.

Mass of sintered glass crucible + nickel complex =  $y$  g.

Mass of dimethyl glyoxime nickel complex =  $(y-x)$  g.

288.69 of nickel complex contain 58.69 g of nickel.

$$\text{Mass of nickel in } (y-x)\text{g of complex} = \frac{(y-x) \times 58.69}{288.69} \text{ g}$$

$$\text{Therefore, Mass of nickel in the whole of the given solution} = \frac{(y-x) \times 58.69 \times 5}{288.69} \text{ g}$$

**Result:**

Mass of Nickel in the given solution = .....g

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